

Hoonki Yeo

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RESEARCH INTERESTS

Optimization under uncertainty; optimization for machine learning; semiparametric modeling.

EDUCATION

Georgia Institute of Technology

Aug 2025 - Dec 2026 (expected)

M.S. in Operations Research, H. Milton Stewart School of Industrial and Systems Engineering

Atlanta, GA

- **Selected coursework:** Convex Optimization, Linear Optimization, Simulation, Theoretical Statistics, High-Dimensional Data Analytics, Computational Data Analytics.

University of Wisconsin-Madison

Aug 2019 - Aug 2022

B.S. in Data Science

Madison, WI

- **GPA:** 3.84/4.00. Graduated with Distinction; Dean's List for four semesters.

RESEARCH EXPERIENCE

Graduate Researcher

Mar 2026 - Present

Georgia Institute of Technology; Advisor: Prof. Sen Na

Atlanta, GA

Equality-Constrained Single-Index Learning via DF-SSQP

- Study derivative-free stochastic optimization for a partially linear single-index model, combining a linear control and a flexible neural link with a unit-norm index constraint.
- Implemented derivative-free stochastic sequential quadratic programming (DF-SSQP) and compared analytic and estimated constraint Jacobians in controlled numerical experiments.
- Investigating whether a learned index of industry-level risk provides incremental predictive information for next-month benchmark volatility beyond current benchmark volatility.

PROFESSIONAL EXPERIENCE

Data Scientist

2023 - 2024

Aivelabs

Seoul, South Korea

- Built production customer-value forecasting workflows, from data pipelines and model training to scoring for retention targeting; used BG/NBD and Gamma-Gamma models for lifetime-value prediction.
- Deployed personalized recommendations for outreach targeting and a multi-armed bandit feedback loop that improved response rate by **6%**.
- Developed customer segmentation using PCA and K-means, and built KPI dashboards to support marketing and sales decisions with cross-functional teams.

Data Analyst Intern

2022 - 2023

Retentics

Seoul, South Korea

- Developed an internal Python preprocessing package that standardized reusable workflows and reduced processing time by **30%**.
- Maintained data-processing and validation pipelines for **10+ clients** and delivered analyses for **5+ clients** to support operational decisions.

TEACHING EXPERIENCE

Peer Tutor

2020 - 2021

University of Wisconsin-Madison

Madison, WI

- Provided one-on-one tutoring in introductory statistics and probability for STAT 301 and STAT 309.

LEADERSHIP & SERVICE

Sergeant, Thermal Observation Device Operator

2016 - 2018

Republic of Korea Army

Yeoncheon, South Korea

- Led a small team responsible for observation-equipment readiness and training new personnel.

TECHNICAL SKILLS

Programming: Python, R, SQL.

Optimization: Gurobi, OR-Tools, linear programming, mathematical modeling.

Scientific computing and modeling: NumPy, PyTorch, scikit-learn, TensorFlow.

Data systems: Apache Airflow, Apache Spark, MySQL, PostgreSQL, Tableau, AWS.

Research tools: Git, LaTeX, Markdown.